Time to Event Models

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In many health sciences applications, binary[1] outcomes[2] are *incompletely observed*. For example, if we are studying whether cancer patients experience a relapse after an initial remission, we may not be able to follow patients to the end of their lives; instead, we may only know whether each patient has relapsed before the end of the study. If a patient has not relapsed by that point, we might not know if they will relapse at some other date or if they will stay cancer-free for the rest of their lives. [3] Their recurrence status at end-of-life is *missing data*. If some study participants withdraw from a study before the scheduled end date, additional missing data arise. Standard logistic regression[4] cannot handle such right-censored[5] data directly without introducing bias[6].

However, these outcome observations are not *entirely* missing. We know that those patients stayed relapse-free *at least* until the time point when they were last observed. If we also know the *time-to-event* for participants who experienced events while under study, we can analyze the *time-to-event-or-study-exit* alongside an event indicator using *survival analysis*. The survival analysis framework provides the foundation for the topics covered in this module.

[1] https://morrison-lab.github.io/sds/data.html#def-binary-var
[2] Intro-to-GLMs.qmd#def-outcome
[3] Binary outcomes are typically defined *for a specific time-point*. It is important to clearly define whether we are interested in outcome status at end of study, at end of life, or at some other time.
[4] logistic-regression.qmd#def-logistic-regression
[5] intro-to-survival-analysis.qmd#def-right-censor
[6] https://morrison-lab.github.io/sds/estimation.html#def-bias